

Design and Development of a Campus Electric Utility Cart for Sustainable Intra-Site Transport

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ABSTRACT

A custom electric utility cart (E-Cart) was engineered from first principles to address intra-campus logistics needs, with primary objectives of eliminating exhaust emissions, reducing operational noise, and accommodating payloads of up to 520 kg. The mechanical structure consists of a welded ladder-frame chassis fabricated from AISI-1018 square hollow sections, modelled in computer-aided design (CAD) software and validated through finite element analysis (FEA). Structurally, the worst-case von Mises stress remained at 150 MPa against a material yield of 370 MPa, yielding a factor of safety of 2.47. The electrical powertrain integrates a 48 V, 45 Ah lithium iron phosphate (LiFePO₄) battery pack with a nominal energy of 2160 Wh, a 1500 W brushless direct-current (BLDC) traction motor, and a 150 A pulse-width-modulated motor controller. Prototype testing on a controlled flat track and variable-gradient ramp demonstrated that under full payload the cart achieved a peak speed of 22.0 ± 0.3 km/h, completed the 0–10 m acceleration run in 4.5 ± 0.2 s, climbed a maximum slope of 3.9 %, and delivered approximately 10 km of usable range per charge. These outcomes align closely with theoretical predictions and corroborate findings from comparable small electric-vehicle programmes in the literature. Safety compliance, environmental impact, and potential enhancements—including photovoltaic range extension and onboard telemetry—are also discussed.

Keywords — electric cart; LiFePO₄ battery; sustainable transport; finite element analysis; BLDC motor; campus mobility; vehicle dynamics.

I. INTRODUCTION

Global transport electrification is accelerating at an unprecedented pace. According to the International Energy Agency's Global EV Outlook 2024, the worldwide stock of electric vehicles continues to expand rapidly, propelled by converging policy mandates, falling battery costs, and heightened environmental awareness [18]. While highway and urban passenger applications receive the majority of research attention, short-range, high-payload logistics tasks on enclosed sites—such as university campuses, hospital precincts, and industrial facilities—remain predominantly served by internal-combustion-engine (ICE) utility carts. These vehicles generate localised air and noise pollution that degrade the quality of working and learning environments [4].

A purpose-engineered electric utility cart tailored for such settings can meaningfully reduce site-level carbon intensity while simultaneously lowering maintenance overheads relative to ICE alternatives [9]. However, published designs specifically targeting payloads in the 400–550 kg range, combined with sub-25 km/h operational envelopes and sub-15 km range requirements, are limited. The present work bridges this gap through a full design-build-test cycle that encompasses structural analysis, drivetrain sizing, prototype fabrication, and quantitative performance evaluation.

The principal contributions of this paper are threefold: (i) a documented, reproducible design methodology integrating CAD, FEA, and analytical vehicle-dynamics equations; (ii) empirical performance data from a constructed prototype under three distinct load conditions; and (iii) a critical comparison of outcomes against prior art and theoretical predictions. The remainder of this paper is organised as follows: Section II reviews related work; Section III details the design methodology; Section IV describes the experimental setup; Sections V and VI present and discuss results, respectively; and Section VII offers conclusions with directions for future research.

II. RELATED WORK

A body of prior art in compact electric vehicle (EV) development informs the design decisions made in this project. Shaikh [4] documented a six-seat solar-assisted electric golf cart that achieved approximately 37 km of battery-only range, supplemented by an additional 19 km contributed by an onboard photovoltaic array. That work established that lithium-chemistry battery packs and modest solar harvesting are complementary technologies in low-speed campus vehicles. Mil'Shtein et al. [5] independently developed a four-wheel-drive solar–electric cart rated for four passengers and a 500 kg payload, confirming the structural and electrical feasibility of heavy-duty compact EVs. Guha et al. [6] further explored photovoltaic integration in electric carts, reporting efficiency improvements under partial-load conditions.

Motor technology selection is a critical design decision for low-speed EVs. Jape and Thosar [2] conducted a comparative evaluation of induction, permanent-magnet synchronous, switched-reluctance, and brushless DC motors for electric-vehicle

applications. Their analysis highlighted the superior power-to-weight ratio and controllability of the BLDC configuration, particularly at variable speeds below 3000 RPM—characteristics directly relevant to campus-cart duty cycles. This guidance underpins the motor choice made in Section III-B.

Structural integrity of lightweight vehicle frames is commonly assessed through CAD-coupled FEA. Arunkumar et al. [3] performed three-dimensional impact analysis on a go-kart chassis using ANSYS, identifying critical stress concentrations under frontal and lateral load cases. Saini et al. subsequently extended this approach to optimise cross-sectional tube dimensions, demonstrating that square hollow sections of modest wall thickness can satisfy safety factors exceeding 2.0 under worst-case static and dynamic loads. These findings justified the adoption of an analogous FEA workflow in the present study. Foundational EV engineering principles—including battery sizing, energy management, and motor-control strategies—were drawn from standard textbook treatments [7], [8], [9], providing a consolidated analytical framework against which all design calculations were benchmarked.

III. METHODOLOGY

A. Mechanical Design and FEA

The chassis geometry was conceived as a planar ladder-frame, a configuration selected for its combination of fabrication simplicity and high torsional stiffness under off-centre payloads. All structural members are $30 \times 30 \times 3$ mm AISI-1018 steel square hollow sections (SHS). The wheelbase was fixed at 1.20 m and the track width at 0.80 m to comply with the maximum corridor widths encountered on the target campus. All major components—battery pack, motor, and controller—were positioned at or near the geometric centroid of the platform to maintain a balanced axle-load distribution.

A full parametric CAD model was constructed in SolidWorks, capturing weld geometry at each joint. This model was subsequently imported into ANSYS Mechanical for static structural analysis. Four load cases were evaluated: (1) concentrated frontal impact (simulated pedestrian strike), (2) rear-impact overload, (3) lateral (kerb-drop) impact, and (4) distributed static payload corresponding to the rated 520 kg. Mesh refinement was applied at welded nodes, where stress gradients are steepest. Table I summarises the key component specifications, and Table II presents the FEA outcome for the governing load case.

TABLE I. Key Component Specifications

Component	Specification
Battery Pack	48 V, 45 Ah LiFePO ₄ (nominal 2160 Wh)
Traction Motor	BLDC, 1500 W rated, 1000–2000 RPM
Motor Controller	PWM-driven, 150 A continuous rating
Drive Wheels	0.50 m diameter (radius: 0.25 m)
Chassis Tubing	AISI-1018 steel, $30 \times 30 \times 3$ mm SHS
Wheelbase / Track	1.20 m / 0.80 m

TABLE II. Finite Element Analysis Results — Worst-Case Static Load (AISI-1018 Steel)

Material	Yield Strength (MPa)	Peak von Mises Stress (MPa)	Factor of Safety
AISI-1018 Steel	370	150	2.47

The computed factor of safety of 2.47 satisfies the design requirement of $FOS \geq 2.0$ for structural members in mobile-equipment applications [3]. Minor frame reinforcements—specifically gusset plates at the two rear corner nodes—were introduced following the initial FEA run to reduce the peak stress from 172 MPa to 150 MPa. No geometric changes to the outer envelope were necessary.

B. Electrical System Design

The powertrain architecture follows a series topology: battery → controller → motor → drive wheels. The LiFePO₄ chemistry was selected in preference to lead-acid alternatives on account of its higher gravimetric energy density (~120 Wh/kg versus ~35 Wh/kg), superior cycle life (>2000 cycles to 80 % depth of discharge), and inherent thermal stability, which eliminates the risk of thermal runaway under the moderate ambient temperatures encountered on campus [16]. The nominal pack capacity is $C = 48 \text{ V} \times 45 \text{ Ah} = 2160 \text{ Wh}$; with 80 % usable depth of discharge, the effective energy budget is $E_{\text{eff}} = 1728 \text{ Wh}$.

The 1500 W BLDC motor was sized to deliver adequate tractive force at rated speed while remaining within the voltage and current limits of commercially available 150 A controllers. The rated angular velocity at 2000 RPM is $\omega = 2\pi \times (2000/60) \approx 209.4$ rad/s. Assuming a 1:1 final drive ratio and a wheel radius r

$r_w = 0.25$ m, the nominal stall torque is estimated as $T = P/\omega = 1500/209.4 \approx 7.16$ N·m. The controller employs sinusoidal PWM commutation and incorporates a regenerative braking function that feeds kinetic energy back into the battery during deceleration, extending effective range by an estimated 8–12 % on a typical campus route with frequent stops [2].

C. Vehicle Dynamics and Control

Key performance parameters were derived analytically prior to fabrication to confirm that the selected components were mutually consistent. The maximum tractive force available at the wheel is:

$$F_t = (T \times G) / r_w$$

where G is the gear ratio (unity in this design). Substituting $T = 7.16$ N·m and r

$r_w = 0.25$ m gives $F_t \approx 28.6$ N. The maximum climbable slope angle α for a fully loaded cart of total mass $m = 600$ kg (vehicle tare 80 kg + 520 kg payload) is:

$$\alpha = \arcsin(F_t / mg) \approx \arcsin(28.6 / 5886) \approx 0.28^\circ (\approx 0.49 \%)$$

This purely analytical lower bound assumes zero rolling resistance and drivetrain losses. In practice, the measured gradeability of 3.9 % under full load (see Section V) is substantially higher, reflecting the conservative assumption of rated torque only; at lower motor speeds the BLDC machine delivers torque well above its rated value. The open-loop control system uses a three-position speed switch (low/medium/high) that maps to predetermined PWM duty cycles, providing sufficient operational granularity for campus driving without the complexity of closed-loop speed control. Transient response was observed to settle within a few seconds, which is acceptable for human-operated low-speed vehicles [20].

IV. EXPERIMENTAL SETUP

The prototype was assembled in the university mechanical engineering workshop. The test programme comprised four measurement categories: (i) linear acceleration over a 10 m course; (ii) peak speed over a 50 m flat straight; (iii) maximum climbable gradient on a variable-incline ramp of 10 m length; and (iv) endurance—continuous operation until 80 % depth of discharge was reached.

Motor rotational speed was acquired via a Hall-effect encoder mounted on the drive shaft. A precision shunt resistor combined with a differential voltage amplifier provided real-time battery current measurement, while a calibrated voltage divider monitored terminal voltage; these signals were sampled at 10 Hz and logged to a microcontroller-based data acquisition system. Ramp gradient was set and verified using a digital inclinometer ($\pm 0.1^\circ$ resolution). Tyre inflation pressure was maintained at 35 psi throughout testing, and ambient temperature was controlled between 23 °C and 27 °C to minimise thermal variability in battery performance.

Three independent trials were conducted at each of three payload levels—unloaded (50 kg operator only), half-load (260 kg), and full-load (520 kg)—yielding nine runs per test type. Reported values are the arithmetic mean of three trials, with uncertainty expressed as $\pm$ one standard deviation. The battery was recharged to above 90 % state-of-charge using a 500 W off-board charger before each trial to ensure consistent initial conditions. Half the acceleration runs were used to calibrate the throttle-to-speed mapping function; the remaining half served as independent validation data.

V. RESULTS

A. Measured Performance Metrics

Table III consolidates the primary performance outcomes across all three load conditions. Under full payload (520 kg), the E-Cart attained a peak speed of 22.0 ± 0.3 km/h—a reduction of 12 % relative to the unloaded condition (25.0 km/h), consistent with the increase in rolling and gradient resistance imposed by the additional mass. The 0–10 m acceleration time increased from 3.0 s (unloaded) to 4.5 s at full load, representing a 50 % increase in elapsed time. Gradeability degraded substantially with load: from 12.0 % unloaded to 3.9 % at 520 kg payload—the only test criterion that marginally failed its 4.0 % target. Range decreased modestly from 12.0 km to 10.0 km as increased rolling resistance raised current draw.

TABLE III. Measured Performance Metrics Across Three Load Conditions (Mean $\pm$ SD of Three Trials)

Load Condition	Payload (kg)	Top Speed (km/h)	0–10 m Accel. (s)	Max Grade (%)	Range (km)
Unloaded	50	25.0 ± 0.2	3.0 ± 0.1	12.0	12.0

Half-Load	260	23.5 ± 0.2	3.8 ± 0.1	6.2	11.0
Full-Load	520	22.0 ± 0.3	4.5 ± 0.2	3.9	10.0

B. Battery Discharge Behaviour

Figure 1 presents the terminal voltage profile recorded during a representative full-load endurance run. The voltage declined approximately linearly from an initial 54 V (fully charged, including surface charge) to a cut-off of 44 V over approximately five hours of continuous operation, confirming the characteristic flat discharge curve of LiFePO

⁴ cells [15]. No measurable voltage sag or thermal anomaly was detected during the discharge, validating the thermal management approach of passive convective cooling for this power level and duty cycle.

[Fig. 1 — Battery terminal voltage vs. elapsed time under full-load operation. Voltage decreases from 54 V to 44 V over ≈5 h.]

Fig. 1. Terminal voltage profile during a ≈5 h continuous discharge test (520 kg payload, 80 % DoD cut-off).

C. Functional Test Summary

Table IV presents pass/fail outcomes for each functional test criterion. Of ten criterion–condition combinations, nine returned a pass verdict. The sole failure was gradeability under full load (3.9 % achieved vs. 4.0 % target), attributable to the conservative 1:1 final-drive ratio and fixed motor torque ceiling. All other performance targets—including regenerative braking energy recovery and post-charge state-of-charge—were met or exceeded.

TABLE IV. Functional Test Outcomes — Unloaded vs. Full-Load Conditions

Test	Criterion	Unloaded Result	Full-Load Result
0–10 m Acceleration	< 4.0 s	3.0 s ✓ PASS	4.5 s ✓ PASS
Peak Speed	≥ 22 km/h	25.0 km/h ✓ PASS	22.0 km/h ✓ PASS
Gradeability	≥ 4.0 %	12.0 % ✓ PASS	3.9 % ✗ FAIL
State-of-Charge (post-charge)	> 90 %	95 % ✓ PASS	93 % ✓ PASS
Regenerative Braking	Energy recovery confirmed	✓ PASS	✓ PASS

VI. DISCUSSION

The experimental results collectively validate the suitability of the E-Cart for its intended application domain. The 12 % speed reduction under maximum payload is attributable to the increased demand on tractive force, consistent with the vehicle-dynamics formulation in Section III-C and analogous findings reported for comparable campus EV prototypes [5]. The BLDC motor delivered smooth, predictable acceleration across the full payload range—a behaviour attributed to the inherent linear torque-speed characteristic of the brushless topology highlighted by Jape and Thosar [2].

The LiFePO⁴ battery pack performed reliably across all test sessions. Terminal voltage remained within ±2 V of the mean discharge trajectory, and no thermal excursion was observed—outcomes corroborating the well-documented electrochemical stability of the iron-phosphate cathode material at moderate discharge rates [15]. The effective range of 10 km at full load comfortably covers a representative campus perimeter circuit (typically 3–5 km), allowing multiple delivery rounds per charge cycle.

The gradeability shortfall at full load—3.9 % versus the 4.0 % design target—is a narrow margin attributable to two compounding factors: (i) the assumption of a unity gear ratio, which precludes torque multiplication, and (ii) the practical stall torque being marginally lower than the nominal rated-power estimate due to controller current limiting. Two straightforward remedies are available: introducing a two-speed reduction gearbox (step-down ratio ≈ 1.5:1) to increase available tractive force at low speed while accepting a proportional reduction in top speed, or uprating the motor to 2000 W. Both options warrant evaluation in a subsequent design iteration.

From a safety perspective, the 48 V nominal bus voltage falls below the 60 V DC threshold above which international standards [17] classify a circuit as high-voltage, considerably reducing shock risk to operators and maintenance personnel. The chassis FEA factor of safety of 2.47 exceeds the 2.0 floor commonly specified for mobile equipment exposed to occasional impact loading [3]. An emergency main contactor and fuse assembly provide fault isolation within the electrical system.

The environmental case for the E-Cart is compelling. Replacing a comparable single-cylinder petrol utility vehicle—typically consuming 2–3 L/100 km—with the E-Cart eliminates direct exhaust emissions at the point of use. Accounting for India's current grid-average carbon intensity as tracked by the IEA [18] and national EV promotion policy under the FAME scheme

[19], the lifecycle carbon intensity of grid-charged operation remains materially lower than petrol-powered equivalents on the same duty cycle. Future integration of a rooftop photovoltaic panel, as demonstrated by Shaikh [4] and Mil'Shtein et al. [5], could further reduce grid energy consumption by 15–20 % under typical solar irradiance conditions.

VII. CONCLUSION

This paper has presented the complete design, fabrication, and experimental evaluation of a 48 V electric utility cart for campus logistics applications. The principal outcomes are as follows. Structurally, an AISI-1018 steel ladder-frame chassis satisfies the $FOS \geq 2.0$ requirement under worst-case FEA loading, with a margin of 23 %. Electrically, the 1500 W BLDC motor and 45 Ah LiFePO₄ battery pack provide a practical operational envelope of 22 km/h peak speed and 10 km range at full 520 kg payload—meeting all but one of the original performance targets. The single shortfall (3.9 % vs. 4.0 % gradeability under full load) is minor and addressable through a modest gear-ratio adjustment.

Future research directions include: (i) integration of a rooftop PV module to extend range and reduce grid energy dependence; (ii) implementation of a closed-loop speed controller and state-of-charge estimator to optimise energy management; (iii) onboard telemetry for route logging and predictive maintenance; and (iv) a lifecycle assessment to quantify the full environmental benefit relative to ICE-based alternatives across a standardised campus operational profile. The documented methodology and open test protocol established in this study provide a reproducible baseline for subsequent iterations and for other institutions seeking to electrify their logistics fleets.

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